

Zhenyu (Bella) Chen

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science–Machine Learning GPA 4.0

Atlanta, GA

Dec 2025

- Relevant Coursework: Deep Learning, Natural Language Processing, Data and Visual Analytics

Georgia Institute of Technology

Bachelor of Science in Computer Science

Atlanta, GA

Dec 2024

- Relevant Coursework: Computer Organization & Program, Computer Vision, Data Structures & Algorithms

EXPERIENCE

AI Evaluation Engineer Intern

Powabase

April 2025 – Present

Remote

- Built an end-to-end Agentic AI evaluation platform that automated knowledge base integration, agent orchestration, benchmark execution, and LLM-based evaluation, supporting 500+ benchmark samples while reducing manual evaluation effort by approximately 90%.
- Integrated Powabase APIs to automate document ingestion, knowledge base creation, and indexing workflows, providing external knowledge sources for downstream agent execution.
- Designed and implemented an agent orchestration workflow to coordinate knowledge retrieval, reasoning, and final answer generation across multiple AI agents, enabling modular and scalable task execution.
- Developed an automated benchmark pipeline to execute hundreds of evaluation tasks, capturing retrieval outputs, reasoning traces, and final responses for reproducible large-scale experiments.
- Implemented an LLM-as-a-Judge evaluation pipeline to automatically assess generated responses, aggregate accuracy metrics, and identify failure cases for workflow optimization.
- Optimized system prompts and reasoning rules through iterative benchmark-driven prompt engineering, improving agent answer accuracy by 30%.

AI Application Engineer Intern

Georgia Tech Infrastructure and Sustainability Department

Aug 2025 – Dec 2025

Atlanta, GA

- Developed an enterprise AI assistant using Microsoft Copilot Studio, delivering 50+ conversational workflows across HR, IT, equipment requests, and other internal service scenarios to automate employee support, business request processing, and enterprise workflows.
- Designed and implemented 50+ workflow topics in Microsoft Copilot Studio by configuring trigger phrases, conversation flows, variables and conditional logic to enable intent-based routing and automate processes.
- Curated domain-specific training data and created JSONL datasets to fine-tune GPT models in Azure OpenAI, improving intent recognition and increasing workflow routing accuracy by approximately 9%.
- Designed structured request forms using Adaptive Card Designer in Microsoft Copilot Studio, configuring form fields for equipment requests, access requests, and other business scenarios, and integrating them with chatbot workflows for seamless user input collection.
- Built automated workflows with Power Automate to validate and route submitted requests to the appropriate SharePoint Lists based on request type and business rules, enabling cross-functional request management for HR, IT, and Facilities teams.

Software Developer Intern

Amazon

July 2023 – Dec 2023

Boston, MA

- Developed a real-time optimization system for Amazon Fulfillment Centers that automatically generated package routing and fluid lane assignment recommendations based on live operational data, streamlining fluid loading operations and reducing end-to-end processing latency by 40%.
- Built an event-driven processing pipeline using AWS Lambda, EventBridge, SQS, and DynamoDB to automate site data ingestion, forecast synchronization, Arc Rate computation, and recommendation updates to DockFlow.
- Developed core backend services in Java 11, implementing operational data processing, Arc prioritization, and business-rule-based fluid lane assignment logic.
- Optimized forecast data access by introducing DynamoDB caching and leveraging Amazon CloudWatch to identify performance bottlenecks, improving system throughput and response time.
- Designed and implemented an automated UI regression testing framework for DockFlowUX, replacing manual regression testing for critical user workflows, reducing UI regression testing time by approximately 70%

- Developed end-to-end automated test cases using Cypress for key business workflows, including the homepage, Workcells, Arcs, and Settings modules, expanding automated test coverage.
- Integrated Cypress into an AWS-based CI/CD pipeline by provisioning the testing infrastructure with AWS CDK, configuring IAM authentication, and enabling automated UI regression testing for every code change prior to deployment.

Software Developer Intern

June 2024 – August 2024

Boke Technology

Shanghai, China

- Developed frontend modules for an internal poker tournament management platform, including tournament management, player management, leaderboard, and match result management using React and TypeScript.
- Integrated RESTful APIs to support tournament creation, player information management, match result queries, and leaderboard updates with dynamic data rendering.
- Built an analytics dashboard with Recharts to visualize tournament participation, player rankings, prize distribution, and tournament performance metrics for operational monitoring.
- Implemented search, filtering, sorting, pagination, and responsive layouts to improve administrative workflow efficiency and overall user experience.

Product Management Intern

Jan 2023 – May 2023

MessageGears

Atlanta, GA

- Authored 15+ Product Requirement Documents (PRDs) and User Stories in Jira, analyzed business workflows, defined functional and non-functional requirements, and participated in technical solution reviews.
- Collaborated with UI/UX designers to create 10+ wireframes and interactive prototypes, translating business requirements into product designs to support requirement reviews and engineering development.
- Coordinated cross-functional collaboration among product, design, and engineering teams, tracking progress across 2–3 Agile sprints, managing release timelines, and driving end-to-end product delivery.
- Conducted Functional Testing and User Acceptance Testing (UAT) for newly released features, identified and tracked defects in Jira, collaborated with engineering teams to validate fixes, and ensured high-quality product releases while supporting continuous product improvements.

TECHNICAL SKILLS

Languages: Java, Python(Pandas, NumPy), SQL, TypeScript, JavaScript, HTML, CSS, R, Powershell

Frameworks: Tableau, Power BI, Excel, Spring (Boot, MVC, Security), React, FastAPI, Node.js, Cypress, JUnit

Libraries: TensorFlow, PyTorch, SciKit-Learn, Pandas, NumPy, OpenCV, Matplotlib

Tools & Platforms: AWS (Lambda, DynamoDB, EventBridge, S3, EC2), Git, Linux, MySQL, PostgreSQL, MongoDB

Concepts: Data Structures, Algorithms, Object-Oriented Design, Machine Learning, Deep Learning, API Design, CI/CD, Statistical Analysis & Visualization

PROJECTS

Financial Reasoning Benchmark for LLM Evaluation (Python)

Sep 2024 – Dec 2024

- Developed a 263-question finance benchmark from open-source finance textbooks to evaluate large language models on domain-specific financial reasoning tasks.
- Designed a GPT-4-based prompt engineering pipeline to automatically generate multiple-choice distractors, followed by manual refinement to improve benchmark quality and question difficulty.
- Built a Python evaluation pipeline to automate benchmark execution, calculate accuracy and error metrics, and analyze model performance across financial reasoning and numerical reasoning tasks.
- Evaluated GPT-4 on the benchmark, achieving 53.2% accuracy with a 17% error rate, providing a standardized dataset for assessing and improving financial-domain LLMs.

Policy Document Analysis System (Python)

Sep 2025 – Dec 2025

- Built an end-to-end AI system for analyzing OECD policy documents, supporting automated policy classification and document-grounded question answering.
- Fine-tuned LLaMA models with LoRA/QLoRA using Hugging Face Transformers to improve domain-specific multi-label classification performance.
- Developed a RAG pipeline by implementing semantic document chunking, Sentence Transformer embeddings, and FAISS-based vector retrieval for long-document reasoning.
- Designed inference and evaluation pipelines to compare fine-tuned LLMs with zero-shot prompting and traditional machine learning baselines.

Interactive Graphical Application – Flyswatter Game (C, Nin10kit, DMA)

Jan 2022 – May 2022

- Developed an interactive Flyswatter game in C with real-time graphics rendering, animation, and score tracking, delivering a smooth gameplay experience.
- Optimized screen rendering using DMA and double buffering (Video Buffer) to reduce flickering and enable smooth object movement.
- Implemented collision detection, object state management, and keyboard input handling to support real-time gameplay and scoring.

Drone Dispatch System (SQL)

Jan 2024 – May 2024

- Developed an end-to-end drone dispatch management system supporting order lifecycle management, delivery scheduling, inventory tracking, and user administration, streamlining logistics operations through a unified desktop platform.
- Designed a normalized relational database and implemented 20+ SQL stored procedures, triggers, and transactions to automate business workflows while ensuring data consistency and integrity.
- Developed a JavaFX desktop application with JDBC-based MySQL integration, delivering CRUD operations, dispatch management, inventory monitoring, and real-time operational reporting.

COVID-19 Case-Fatality Ratio Visualization (Java)

Oct 2020 – Dec 2020

- Developed a Java desktop application to process and visualize COVID-19 case-fatality ratio data across 5 racial groups and 6 U.S. states, enabling interactive comparison through bar chart visualizations.
- Implemented CSV parsing, data preprocessing, and statistical aggregation using Java Collections and custom Doubly Linked List data structures to support efficient sorting and filtering.
- Built a GUI with interactive controls for selecting states and demographic groups, improving data exploration and visualization usability.